Core Focus on Proportions & Probability

Block 3 Review ~ Percents

ANSWER KEY

Name _____ Period _____ Date _____

Part I – Selected Response

1. Marco wrote some pairs of numbers on the board and said they were equal. Decide which pairs really are equal. Circle all that apply.

A. 0.4 = 4%

B. 1.5 = 150%

C. 0.02 = 2%

D. $\frac{7}{10}$ = 7%

E. $\frac{3}{8}$ = 37.5%

F. $\frac{1}{2}$ = 0.5%

B, C, E. (A: 0.4 = 40%. D: 7/10 = 70%. F: 1/2 = 50%.)

2. What is 40% of 85?

A. 3.4

B. 34

C. 45

D. 340

B. 0.4 × 85 = 34

3. Fifteen is 5% of what number?

A. 0.75

B. 3

C. 300

D. 750

C. 15 = 0.05x → x = 15 ÷ 0.05 = 300

4. What percent of 50 is 70?

A. 14%

B. 71%

C. 120%

D. 140%

D. 70 ÷ 50 = 1.4 = 140%

5. What is the percent decrease from 60 to 45?

A. 15%

B. 25%

C. $33\frac{1}{3}$%

D. 75%

B. Change = 15; 15 ÷ 60 = 0.25 = 25%

6. Last month, 25 kids came to chess club. This month, 35 came. What is the percent increase?

A. 10%

B. 28.6%

C. 40%

D. 140%

C. Change = 10; 10 ÷ 25 = 0.40 = 40%

7. Ava bought a lamp for $80. Her town has a 5% sales tax. How much money did Ava pay in sales tax?

A. $0.40

B. $1.60

C. $4.00

D. $40.00

C. 0.05 × 80 = $4.00

8. **Noah bought new sneakers. The original price was $80, but they were on sale for 35% off. For numbers 8a – 8d, determine whether each statement about Noah's purchase is true or false.**

8a. Noah paid less than $80 for the sneakers. **TRUE FALSE**

8b. The discount was $35. **TRUE FALSE**

8c. Noah paid $108 for the sneakers. **TRUE FALSE**

8d. Noah paid $52 for the sneakers. **TRUE FALSE**

8a TRUE · 8b FALSE (0.35 × 80 = $28) · 8c FALSE ($108 adds the discount) · 8d TRUE ($80 − $28 = $52)

Form C

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Part II – Constructed Response

Write each decimal or fraction as a percent.

1. 0.65

65% (0.65 × 100)

2. $\frac{4}{5}$

80% (4/5 = 80/100)

Solve. Show all work necessary to justify your answer.

3. What is 60% of 150?

90 ($x = 0.6 \times 150$)

4. Eighteen is 12% of what number?

150 ($18 = 0.12x \rightarrow x = 18 \div 0.12$)

5. What percent of 30 is 21?

70% ($21 = x \cdot 30 \rightarrow x = 0.7$)

6. Find the percent of decrease from 50 to 42.

16% decrease (change 8; 8 ÷ 50 = 0.16)

Find each selling price. Use mathematics to justify your answer.

7. original price: $45.00
percent of markup: 20%

$54.00 (markup 0.2 × 45 = $9; $45 + $9)

8. original price: $70.00
percent of discount: 30%

$49.00 (discount 0.3 × 70 = $21; $70 − $21)

9. A $180 guitar was on sale for 15% off the original price. How much money does a person save if she buys it on sale? Show all work necessary to justify your answer.

$27 saved (0.15 × 180 = 27)

10. The swim team raised $1,200 last year. This year the team hopes to raise 30% more money. How much money does the team hope to raise this year? Show all work necessary to justify your answer.

$1,560 (0.3 × 1,200 = $360 more; $1,200 + $360 = $1,560)

11. Priya bought a new bike. The bike cost $600. Her state has a 7.5% sales tax. Priya did the work shown below to figure out how much she would pay for the bike, including the sales tax. Show another way to find the total amount Priya paid for the bike.

Let x = tax to pay

$x = 600 \cdot 0.075$
$x = 45$
I will pay $45.00 in tax.

Total cost for the bike:
$600 + $45.00 = $645.00

I will pay $645.00 for the bike.

$645.00. Sample: she pays 100% + 7.5% = 107.5% of the price, so 1.075 × 600 = $645.00. Or a proportion: 107.5/100 = x/600 → x = 645.

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Part III – Problem Solving

A couch was originally $400. It was on sale for 40% off the original price. When the Garcia family bought the couch, they found out it was an additional 25% off the sale price. What was the overall percent of discount from the original price of the couch to the price the Garcias paid?

Sale price: 0.6 × 400 = $240. Then 25% off: 0.75 × 240 = $180. Savings: $400 − $180 = $220. Overall discount: 220 ÷ 400 = 0.55 = **55%** (not 65%). Check: 0.6 × 0.75 = 0.45, so they paid 45% of the original.